Shantanu Naresh Sonawane

Pune, Maharashtra, India
sonawaneshantanu1012@gmail.com | linkedin.com/in/shantanu-sonawane | shantanusonawane25.github.io/Portfolio

Education

Sinhgad College of Engineering Pune, Maharashtra
Bachelor of Engineering in Information Technology *Expected May 2027*

- Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Object-Oriented Programming, Web Technologies, Data Science, Big Data Analysis

Technical Skills

Languages: Java, Python, C/C++, JavaScript, SQL, Bash, HTML/CSS
Frameworks & Libraries: React.js, Node.js, Express.js, Flask, scikit-learn, REST APIs
AI/ML: NLP, TF-IDF text classification, Retrieval-Augmented Generation (RAG), embeddings, vector search (FAISS)
Databases: PostgreSQL (Supabase), MySQL, MongoDB
Developer Tools: Git, GitHub, Docker, pytest, FFmpeg, Linux, VS Code

Projects

AURORA: Music Streaming Platform | *React 19, Vite, Node.js, Supabase (Auth, PostgreSQL, Storage), FFmpeg*

- Built a full-stack streaming app with authentication, user profiles, follows, playlists, liked songs, search, and a player with queue, seek, volume, shuffle and repeat
- Designed a Node.js upload service using FFmpeg/FFprobe for metadata analysis and cover art, supporting single and bulk MP3 uploads with separate upload and publish steps
- Implemented idempotency keys and retry protection to prevent duplicate and stale uploads, covered by 88 automated tests
- Building an artist submission and admin moderation workflow with role-based access and row-level security

Fake Product Review Detector (In Progress) | *Python, scikit-learn, Flask, React.js*

- Building a text classifier in Python (scikit-learn, TF-IDF features) to flag likely fake product reviews using a public labelled review dataset
- Serving the model through a Flask REST API that returns a trust score for a submitted review
- Adding a React.js front end to paste a review and view its score, with pytest unit tests and a Dockerfile to run it

Document Q&A Assistant using RAG (In Progress) | *Python, FAISS, Flask, React.js*

- Building a retrieval-augmented generation (RAG) assistant that answers questions over uploaded PDF documents and cites the source passage for each answer
- Implementing document chunking, embedding generation and top-k retrieval from a FAISS vector index to ground LLM responses
- Evaluating retrieval and answer quality on a set of test questions
- Serving the assistant through a Flask REST API with a React.js chat interface

Leadership & Activities

Student Developer September 2023 – Present
Personal Projects & Technical Development

- Building full-stack web applications with React.js, Node.js, Express.js, PostgreSQL (Supabase), MySQL and MongoDB
- Writing automated tests (pytest, 88-test upload service suite) and using Git/GitHub for version control across projects
- Applying data structures, operating systems and database fundamentals to design reliable backend services

Key Highlights

- Strong foundation in data structures, algorithms, operating systems and object-oriented design
- Experience building end-to-end web applications with React.js, Node.js, REST APIs and SQL/NoSQL databases
- Hands-on machine learning and LLM applications: text classification and retrieval-augmented generation (RAG)
- Comfortable with Linux, Bash scripting and Git/GitHub workflows